Metrics Summary & Feature Importance

Random Forest & Gradient Boosting | Chronological Test Set 2020–2022

Full Metrics (from metrics.json)

Metric	Mean Baseline	Median Baseline	Ridge	Random Forest	GBR
MAE ($)	5,125	3,083	4,391	3,066	3,124
RMSE ($)	12,305	12,616	11,991	11,326	11,279
R^2	-0.002	-0.053	0.048	0.151	0.158
MedAE ($)	3,598	238	2,811	1,123	1,316

Train size: 21,622 (2011–2019) | Test size: 6,269 (2020–2022)

Top 15 Features – Random Forest Importance

Rank	Feature	Importance
1	AGE_AT_START	0.211
2	GENDER_M	0.134
3	ENCOUNTERCLASS_urgentcare	0.091
4	GENDER_F	0.072
5	ENCOUNTERCLASS_ambulatory	0.066
6	PAYER_NAME_NO_INSURANCE	0.059
7	DOW	0.056
8	RACE_white	0.042
9	ETHNICITY_hispanic	0.037
10	RACE_native	0.037
11	ETHNICITY_nonhispanic	0.036
12	ENCOUNTERCLASS_inpatient	0.034
13	MONTH	0.023
14	YEAR	0.022
15	RACE_asian	0.021

Age at encounter start is the strongest single predictor, followed by gender and specific encounter classes. The NO_INSURANCE indicator ranks highly, reflecting systematic cost/coverage differences. Calendar features (day-of-week, month, year) contribute modestly.

Notes on Interpretation

Feature importances are from the Random Forest impurity-based measure after one-hot encoding. They indicate relative contribution within this model and this synthetic dataset; they should not be interpreted as causal effects. Because procedure-level drivers were excluded to avoid leakage, the model captures demographic, payer, and encounter-class patterns only.